A COUNTEREXAMPLE TO WONG'S CONNECTEDNESS-PRESERVING BIJECTION PROBLEM

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Abstract. We construct a bijection of the Euclidean plane that sends every connected subset to a connected subset, while its inverse sends a compact nondegenerate arc to a Cantor set. This gives the answer `False` to the universal statement of Wong's connectedness-preserving bijection problem, MathOverflow 235893, in the Formal Conjectures library. The construction follows the recurrent-tube strategy of L. Peter. Explicit ambient shears produce a tube whose whole cross-sections recur near every point of its complement. Colored subarcs of a retained boundary ray supply disjoint dense connected sets for a prescribed bijection on a singular source ray. A separation argument then proves connectedness preservation for arbitrary connected sets. The complete negative answer is kernel checked in a single Lean 4 source file, using the repository's original predicate and no additional existence axiom.

1. Introduction

A map is *connectedness preserving*, or *Darboux*, if it sends every connected subset of its domain to a connected subset of its codomain. Wong asked whether a Darboux bijection $f : \mathbb{R}^n \to \mathbb{R}^n$, for $n > 1$, must have a Darboux inverse [4]. The Formal Conjectures repository records the corresponding universal assertion [2]. Related continuity results for Darboux injections between manifolds are discussed by Banakh and Banakh [1].

Theorem 1.1. *There is a bijection $f : \mathbb{R}^2 \to \mathbb{R}^2$ that preserves connectedness of every connected subset, together with a compact nondegenerate arc $A \subset \mathbb{R}^2$, such that*

$$f^{-1}(A) = C_0,$$

where C_0 is the middle-thirds Cantor set in $[0, 1] \times \{0\}$. Consequently, f^{-1} does not preserve connectedness.

The recurrent-tube construction and the use of colored recurrent subarcs appear in L. Peter's September 2026 preprint [3]. His latest version already claims a negative answer in every dimension $n \geq 2$. The singular-ray interval-density device is credited there to Gro-Tsen's MathOverflow answer [4]. We make no priority claim for these mathematical ideas. Our contribution here is an explicit planar implementation by ambient shears and a complete Lean verification of the negative answer to the exact repository statement. The proof below is self-contained and requires no prime-end or planar embedding theorem.

Throughout, connected sets are nonempty. We use ordinary classical choice for set bijections; no continuum hypothesis is assumed. Write $E = [0, \infty) \times \{0\}$ for the singular source ray and $\mathfrak{c} = |\mathbb{R}|$ for the cardinality of the continuum.

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2. The recurrent-level argument

Lemma 2.1. *Let X, Y be topological spaces, let $\rho : X \to [0, \infty)$ be continuous, and put $Z = \rho^{-1}(0)$. Suppose $f : X \to Y$ satisfies:*

1. *f is continuous on $X \setminus Z$;*
2. *f preserves connectedness on connected subsets of Z;*
3. *for every $y \in Z$, every open neighborhood O of $f(y)$, and every $\varepsilon > 0$, there exists $\tau \in (0, \varepsilon)$ with*
$$f\big(\rho^{-1}(\tau)\big) \subset O.$$

Then f preserves connectedness on every connected subset of X.

Proof. Let $C \subset X$ be connected. The assertion follows from (1) or (2) if C lies entirely outside Z or entirely inside Z. Otherwise suppose $f(C)$ has a separation by open sets $U, V \subset Y$: they cover $f(C)$, both meet $f(C)$, and $f(C) \cap U \cap V$ is empty. There exist $x \in C \setminus Z$ and $y \in C \cap Z$ whose images lie on opposite sides. Indeed, if an initial such pair lies on the same side, a point of C mapping to the other side can replace one of the pair. Interchange U, V so that $f(x) \in U$ and $f(y) \in V$.

Choose $0 < \tau < \rho(x)$ with $f(\rho^{-1}(\tau)) \subset V$. Define

$$G = \{z \in C : \rho(z) > \tau,\ f(z) \in U\}.$$

This is relatively open in C, since f is continuous wherever $\rho \neq 0$. Using the covering and disjointness of the separation, and the prescribed image of the level τ, its relative complement is

$$C \setminus G = \{z \in C : \rho(z) < \tau\} \ \cup\ \{z \in C : \rho(z) \neq 0,\ f(z) \in V\}.$$

This is relatively open as well. Thus G is clopen in C, contains x, and omits y, a contradiction. No compactness or path assumption on C is used. □

3. A finite deformation by ambient shears

For $H > 0$, put $R_H = [-1, 1] \times [0, H]$. We first establish the finite operation from which the entire tube will be assembled.

Lemma 3.1. *Let $0 < H < T$, and let $O \subset \mathbb{R}^2$ be open with $O \setminus R_H \neq \varnothing$. There is a homeomorphism $g : \mathbb{R}^2 \to \mathbb{R}^2$ fixing R_H pointwise such that*

$$g([-1, 1] \times \{T\}) \subset O.$$

Proof. For a continuous $a : \mathbb{R} \to \mathbb{R}$, the horizontal and vertical shears

$$(x, y) \longmapsto (x + a(y), y), \qquad (x, y) \longmapsto (x, y + a(x))$$

are homeomorphisms: their inverses subtract the same function. Write $s_+ = \max(0, s)$ and define

$$u(y) = \frac{(y - H)_+}{T - H}, \qquad v_X(x) = \frac{(|x| - 1)_+}{|X| - 1} \quad (|X| > 1).$$

The first function vanishes on $[0, H]$ and equals 1 at T; the second vanishes on $[-1, 1]$ and equals 1 at X.

Choose $(X, Q) \in O \setminus R_H$. If $Q \in [0, H]$, then $|X| > 1$. The horizontal shear $a = Xu$, followed by the vertical shear $a = (Q - T)v_X$, fixes R_H and sends $(0, T)$ to (X, Q). If $Q \notin [0, H]$, first use the horizontal shear $a = 2u$ and the vertical shear $a = (Q - T)v_2$. The resulting point is $(2, Q)$. For the final horizontal shear use $a = (X - 2)w$, where

$$w(y) = \begin{cases} (-y)_+/(-Q), & Q < 0, \\ (y - H)_+/(Q - H), & Q > H. \end{cases}$$

Again each shear fixes R_H, and their composition h sends $(0, T)$ to (X, Q).

The open set $h^{-1}(O)$ contains $(0, T)$, hence contains a ball of some radius $\delta > 0$ in the maximum norm. Set $a = \min(\delta/2, 1/2)$, $b(y) = \min(1, u(y))$, and $c(y) = 1 - b(y) + ab(y)$. Then $0 < a \le c(y) \le 1$. The map

$$s(x, y) = (c(y)x, y)$$

is a homeomorphism, fixes R_H, and sends the whole cap at height T into that ball, since $c(T) = a < \delta$. Take $g = h \circ s$. □

4. The recurrent infinite tube

Let $(B_i)_{i\ge 0}$ be a countable basis of open sets in $\mathbb{R}^2$. Choose a schedule $\sigma : \mathbb{N} \to \mathbb{N}$ in which every index occurs arbitrarily late; the first coordinate of the inverse of the standard pairing bijection suffices. Put $H_n = e^n$ and $R_n = R_{H_n}$. Recursively construct ambient homeomorphisms h_n, starting with $h_0 = \mathrm{id}$. If

$$h_n^{-1}(B_{\sigma(n)}) \setminus R_n \neq \varnothing,$$

apply Lemma 3.1 to choose g_n fixing R_n and sending the next cap into this preimage. Otherwise take $g_n = \mathrm{id}$. Set $h_{n+1} = h_n \circ g_n$. Thus

$$h_m|_{R_n} = h_n|_{R_n} \qquad (m \ge n). \tag{1}$$

Write $S = [-1, 1] \times [0, \infty)$ and $P = (-1, 1) \times (0, \infty)$. Define $F : S \to \mathbb{R}^2$ by $F(z) = h_n(z)$ whenever $z \in R_n$; equation (1) makes this well-defined. Every point of S has a neighborhood in S lying below some H_n, where F agrees with h_n. Consequently F is continuous. Two points lie in a common R_n, so injectivity of h_n also proves injectivity of F.

Put

$$U = F(P) = \bigcup_{n\ge 0} h_n(\operatorname{int} R_n), \qquad K = \mathbb{R}^2 \setminus U.$$

The set U is open, and $F|_P : P \to U$ is a homeomorphism. Indeed, for an open subset V of P, its image is the union of the open sets $h_n(V \cap \operatorname{int} R_n)$. The injectivity of F further implies

$$F(S \setminus P) \subset K. \tag{2}$$

Proposition 4.1. *For every $p \in K$, every open neighborhood O of p, and every $m \in \mathbb{N}$, there is $j \ge m$ such that*

$$F([-1, 1] \times \{e^j\}) \subset O.$$

Proof. Choose B_i with $p \in B_i \subset O$. For any n, this basis set cannot be contained in $h_n(R_n)$: being open, it would put p in $\operatorname{int} h_n(R_n) = h_n(\operatorname{int} R_n) \subset U$. Thus $h_n^{-1}(B_i) \setminus R_n$ is nonempty. At any sufficiently late occurrence $\sigma(n) = i$, the construction puts the full next cap in B_i. Stabilization gives the assertion with $j = n + 1$. □

Fix a homeomorphism $a : \mathbb{R} \to (-1, 1)$ and set

$$\Phi(u, v) = F(a(u), e^v).$$

This is a homeomorphism $\mathbb{R}^2 \to U$. Proposition 4.1 implies

$$\forall p \in K\ \forall O \ni p \text{ open } \forall m \in \mathbb{N} \quad \exists j \ge m : \quad \Phi(\mathbb{R} \times \{j\}) \subset O. \tag{3}$$

The retained lateral ray $r(t) = F(1, t)$, $t \ge 0$, is continuous and injective, lies in K, and every tail is dense in K. For the last assertion, choose arbitrarily large j in Proposition 4.1 and use the endpoint $r(e^j)$ of the cap. The bottom arc

$$A = F([-1/2, 1/2] \times \{0\}) = [-1/2, 1/2] \times \{0\}$$

is a compact nondegenerate arc in K, disjoint from the lateral ray. The equality holds because $h_0 = \mathrm{id}$ and all later charts fix R_0. Disjointness follows from injectivity of F.

5. Disjoint dense connected reservoirs

We construct pairwise disjoint compact subarcs J_n of $r([0,\infty))$ such that J_n meets every B_i, $i \le n$, which meets K. Choose disjoint closed parameter intervals successively. After the previous interval ends, start the next one with a positive gap. Tail density of r supplies a parameter in each of the finitely many required basis sets. Choose the new upper endpoint larger than all these parameters and the lower endpoint. This gives the required nondegenerate subarc J_n. All J_n lie in K and are disjoint from A.

Partition $\mathbb{N}$ into countably many pairwise disjoint infinite sets N_c, $c \ge 0$, using $N_c = \{\mathrm{pair}(c,m) : m \ge 0\}$. Define

$$D_c = \bigcup_{n \in N_c} J_n.$$

Each N_c is unbounded. Hence each D_c is dense in K: given $p \in K$ and $p \in B_i \subset O$, any $n \in N_c$ with $n \ge i$ provides a point of $J_n \cap B_i$.

Lemma 5.1. *Each D_c is connected and has cardinality $\mathfrak{c}$. The sets D_c are pairwise disjoint and disjoint from A.*

Proof. For connectedness, suppose open sets V, W separate D_c. Choose points $p \in D_c \cap V$ and $q \in D_c \cap W$, and basis sets $p \in B_i \subset V$, $q \in B_j \subset W$. A sufficiently large $n \in N_c$ satisfies $n \ge i, j$. Then the connected arc $J_n \subset D_c$ meets both V and W, contradicting the separation. Each D_c contains a nondegenerate arc and is a subset of the plane, so its cardinality is $\mathfrak{c}$. The remaining assertions follow from disjointness of the parameter intervals and of A from r. □

6. The bijection on the singular ray

Reserve D_0 for completion of the target bijection. On E, let C_0 be the standard Cantor set, and reserve a translated Cantor set $C_* \subset [2,3] \times \{0\}$. Enumerate all nonempty rational open intervals $I_n \subset (0,\infty)$. Choose successively compact Cantor sets $C_{n+1} \subset I_n \times \{0\}$ disjoint from $C_*, C_0, \ldots, C_n$. This is possible because the finite union of the excluded compact Cantor sets is nowhere dense, leaving a smaller interval in which a scaled Cantor set can be placed.

Choose bijections $C_0 \to A$ and $C_{n+1} \to D_{n+1}$. The remaining part of E contains C_* and the remaining part of K contains D_0; each has cardinality $\mathfrak{c}$. Complete the partial map by a bijection between these remaining parts. We obtain a bijection $g : E \to K$ satisfying

$$g^{-1}(A) = C_0, \qquad g(C_{n+1}) = D_{n+1}. \tag{4}$$

The map g preserves connectedness on E. A connected subset of E is a singleton or an interval. Every nondegenerate interval contains some rational interval I_n, and therefore contains C_{n+1}. Its image T satisfies

$$D_{n+1} \subset T \subset K \subset \operatorname{cl} D_{n+1}.$$

Any set between a connected set and its closure is connected. The singleton case is immediate.

7. Slit-plane coordinates and the final bijection

For $z = (x,y) \in \mathbb{R}^2$, define the continuous nonnegative function

$$\rho(z) = \sqrt{\frac{\sqrt{x^2+y^2} - x}{2}}.$$

It vanishes precisely on E. On $\mathbb{R}^2 \setminus E$, the map

$$\Psi(x,y) = \left(\frac{y}{2\rho(x,y)}, -\log \rho(x,y) \right) \tag{5}$$

is a homeomorphism to $\mathbb{R}^2$, with inverse

$$(u,v) \longmapsto \left(u^2 - e^{-2v},\ 2ue^{-v}\right).$$

These identities follow by squaring and using $\rho > 0$ off the ray.

Proof of Theorem 1.1. Define

$$f(z) = \begin{cases} g(z), & z \in E, \\ \Phi(\Psi(z)), & z \notin E. \end{cases}$$

The two pieces are bijections $E \to K$ and $\mathbb{R}^2 \setminus E \to U$, respectively, so f is a bijection of the plane. It is continuous off E, and its restriction to E preserves connectedness by Section 6.

Let $y \in E$ and let O be an open neighborhood of $f(y) \in K$. Given $\varepsilon > 0$, choose m so large that $e^{-m} < \varepsilon$. By (3), some $j \geq m$ satisfies $\Phi(\mathbb{R} \times \{j\}) \subset O$. Put $\tau = e^{-j}$. Equation (5) sends the whole level $\rho^{-1}(\tau)$ into $\mathbb{R} \times \{j\}$, so $f(\rho^{-1}(\tau)) \subset O$. Lemma 2.1 now proves connectedness preservation for every connected subset of the plane.

Finally, $A \subset K$, and (4) gives $f^{-1}(A) = C_0$. The arc A is connected whereas C_0 is disconnected. For example, $1/4, 3/4$ belong to the Cantor set and $1/2$ does not; a connected subset of the real line containing the first two would contain the third. Thus the inverse of f fails to preserve connectedness. □

8. The exact repository statement and Lean verification

The library's predicate is

$$\text{IsConnectedMap}(f) \quad \Longleftrightarrow \quad \forall C,\ \text{IsConnected}(C) \Longrightarrow \text{IsConnected}(f(C)).$$

The assertion in its MathOverflow 235893 file is

$$\forall n > 1\ \forall (f : \mathbb{R}^n \simeq \mathbb{R}^n), \quad \text{IsConnectedMap}(f) \Longrightarrow \text{IsConnectedMap}(f^{-1}), \tag{6}$$

where $\simeq$ means a set bijection. Theorem 1.1 at $n = 2$ refutes this universal assertion.

All custom mathematical proofs are consolidated in `Wong235893.lean`. Its only imports are Mathlib and the unmodified repository statement module. The construction uses `IsConnectedMap` from namespace `Mathoverflow235893`. Its final theorem is `wong235893_repository_answer`, an unconditional equivalence between `False` and assertion (6). The file also exposes the bijection as `wongCounterexample`, proves its forward preservation and inverse failure, and verifies the compact nontrivial connected witness with its Cantor preimage.

A compiler check reads the original theorem declaration's type and requires its right-hand side to be definitionally equal to the target in this file. It separately checks the final unconditional answer type. The repository's admitted theorem proof is never used. The complete source is checked with Lean 4.33.1, Mathlib revision `0df444a`, and Formal Conjectures revision `89294ea`. Implicit variable creation is disabled and warnings are fatal.

An axiom audit covers every declaration in the custom namespace and allows only Lean's standard axioms `propext`, `Classical.choice`, and `Quot.sound`. In particular, the unconditional answer depends on no `sorryAx`, extra axiom, or unproved geometric existence assumption. The older inverse-limit and composant developments retained in the source are auxiliary results; the final proof uses the direct recurrent tube constructed above.

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References

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